

SUNCLAVE · REQUIREMENTS

SunClave requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/sunclave		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with design case and status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

SunClave requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see SCL-PRB-001). The status column gives the first-order estimate from SCL-PRC-001; every status is an estimate.

The **design case** is a clear day with 700 W/m² DNI, 25 °C ambient, wind below 3 m/s, a site near sea level, and a load of 2.0 kg of unwrapped or single-wrapped solid stainless instruments in a 0.5 kg basket, with 1.5 L of water in a 12 L aluminum cooker.

Safety: These requirements describe a pressure vessel heated by concentrated sunlight. R8 to R11 are safety requirements and take precedence over throughput and cost.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status (estimate)
R1	Sterilizing condition	Saturated steam at 121 °C or more, and not above 124 °C, held for 20 min (unwrapped) or 30 min (wrapped), measured in the load zone	Calculation; later instrumented cycles with biological indicators	Met near sea level; not met above about 300 m with a 103 kPa cooker (about 118 °C at 1,800 m)
R2	Load capacity	Basket at least 250 mm diameter by 150 mm deep; 2.0 kg of solid instruments per cycle (about two delivery kits or four suture kits)	Model check	Met by design (270 x 180 mm basket)
R3	Load type	Solid instruments only, unwrapped or single-wrapped in porous wrap; no lumened, hollow or textile loads	Instructions and design review	Met by definition; a limit of the concept
R4	Cycle time	Cold start to end of hold in 90 min or less in the design case	Energy calculation; later timed cycles	Met on estimate (about 55 to 90 min); at risk , since Kaseman et al. measured 93 to 183 min heat-up with a larger vessel
R5	Daily throughput	3 or more complete cycles on a clear day (DNI 600 W/m ² or more from 09:00 to 15:00 solar time)	Energy calculation	Met on estimate (about 3); zero on overcast days
R6	Cycle record	Load-zone temperature and chamber pressure logged at 1 s intervals from start to end; cycle ID, date and time, hold duration, minimum hold temperature and a pass or fail flag stored for 1,000 or more cycles and shown to the operator at cycle end	Design review; later bench test	Met by design
R7	Measurement accuracy	Temperature within 0.5 °C at 121 °C; pressure within 5 kPa; field check against boiling water at each site	Component datasheets; calibration procedure at TRL 3	Met on datasheets (Pt100 class A about 0.4 °C); pressure about 5 kPa, marginal for R12
R8	Pressure safety	Working pressure 103 kPa (15 psi) gauge set by the cooker's regulator; an independent relief valve set at 125 kPa gauge or less; the cooker's own overpressure plug retained; vessel rated by its maker for the working pressure	Design review	Met by design
R9	Heat input control	Operator can stop heat input within 10 s by turning the dish off the sun, and the dish is shaded when parked	Design review	Met by design
R10	Boil-dry protection	At least 0.5 L of water left at the end of a cycle in the design case; logger alarm if base temperature exceeds 140 °C	Water budget; later test	Met on estimate (about 1.0 L left) and by design (base thermocouple in the logger)
R11	Burn and glare protection	No exposed surface above 60 °C within reach during normal operation other than the lid and	Design review	Partly met; guard not yet defined

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status (estimate)
		handles; focal zone guarded; eye protection supplied		
R12	Air removal check	Logger flags any hold where measured temperature is more than 2 °C below the saturation temperature for the measured pressure	Calculation; later test with deliberate air leak	Met by design; resolution limited by R7 pressure accuracy
R13	Off-grid logger power	Logger runs 3 or more days with no sun on its own battery and recharges from its own panel	Power budget	Met on estimate (about 5 days)
R14	Tracking effort	Retarget no more often than every 15 min; aim by a shadow gnomon without looking at the sun or the focus	Optics estimate	Met on estimate
R15	Stability	Does not tip in a 10 m/s wind with the dish at any tilt and the cooker loaded	Moment calculation	Met on estimate, thin margin (ballast may be needed)
R16	Portability and build	Total mass 40 kg or less; separates into pieces of 20 kg or less; built with hand tools, a drill and bolted joints	Mass estimate; model check	Met on estimate (about 33 kg)
R17	Cost	Parts for one prototype \$400 or less, validation consumables excluded	Priced BOM (bom/ bom.csv)	Not met: about \$437 (about 9 % over)

■ Requirements not met or at risk

- **R1 at altitude (not met).** A cooker regulated at 103 kPa above ambient reaches 121 °C only below about 300 m. See the options in SCL-PRC-001.
- **R17 cost (not met).** About \$437 against \$400.
- **R4 cycle time (at risk).** The estimate is well below the only measured precedent.
- **R11 guard (not yet in the concept).** A focal-zone guard needs to be defined at TRL 3.

■ Assumptions

- Dish aperture 1.4 m (1.54 m²), overall optical efficiency about 73 % (reflectance, spillage and shading), pot base absorptance about 0.9.
- Saturated steam temperatures from the Antoine equation for water; standard atmosphere for ambient pressure with altitude.
- Hold times follow CDC Table 7 and the IARC manual cited in SCL-PRB-001. Equivalent hold at lower temperature uses a z-value of 10 °C, which is a common engineering assumption and not a validated claim.
- "Pass" in R6 means the recorded cycle met R1 and R12. It does not prove sterility; biological and chemical indicators remain the reference.